Note 5.2 Orthogonal Subspaces

Intro

Property of $m \times n$ Matrix A

Each vector in $N(A)$ is orthogonal to every vector in column space of A^T. That is,

$$R(A^T) \perp N(A)$$

Proof can be done by the definition of null space and the multiplication of A and $\mathbf{x}$.

Definition of Orthogonal Subspaces

Orthogonal for every vectors in two subspaces.

Note that sometimes orthogonal subspaces are not agree with intuitive idea of perpendicularity.

Definition of Orthogonal Complement

The set of all vectors that are orthogonal to every vector for a subspace of $\mathbb{R}^n$. Denoted by $Y^\perp$.

Remarks

1. Two orthogonal subspaces have only common elements $\mathbf{0}$.
2. The orthogonal complement of a subspace of $\mathbb{R}^n$ is also a subspace of $\mathbb{R}^n$.

The two remarks can be proved by definition of inner product and orthogonality.

Fundamental Subspaces

Intro

The column space of a matrix is the same as its range.

Theorem 5.2.1 Fundamental Subspaces Theorem

For a $m \times n$ matrix A,

$$N(A) = R(A^T)^\perp, N(A^T) = R(A)^\perp$$

The proof can be done by the property in Intro.

Theorem 5.2.2

The sum of the dimension of a subspace of $\mathbb{R}^n$ and its orthogonal complement is n. That is

$$\dim S + \dim S^{\perp} = n$$

And the conjunction of their basis is the basis of $\mathbb{R}^n$.

It can be proved by theorem 5.2.1.

It is also prove that for each vector in $\mathbb{R}^n$ can be expressed uniquely as a sum of two vectors respectively in a subspace of $\mathbb{R}^n$ and its orthogonal complement.

Definition of Direct Sum

For each vector $\mathbf{w} \in W$, it can be expressed uniquely as a sum $\mathbf{u} + \mathbf{v}$ where $\mathbf{u} \in U$ and $\mathbf{v} \in V$. And W is a direct sum of U and V, denoted by $W = U \oplus V$.

Theorem 5.2.3

The direct sum of a subspace in $\mathbb{R}^n$ and its orthogonal complement equals to $\mathbb{R}^n$.

Theorem 5.2.4

For a subspace S of $\mathbb{R}^n$, $(S^{\perp})^{\perp} = S$.

Corollary 5.2.5

For a $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$, either there is a vector $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$ or there is a vector $\mathbf{y} \in \mathbb{R}^m$ such that $A^T\mathbf{y} = \mathbf{0}, \mathbf{y}^T\mathbf{b} \neq 0$.

(That is, $\mathbf{b}$ in the column space of A or not)

And the matrix A establish a one-to-one correspondence between $R(A^T)$ and $R(A)$.

$$R(A) = \{A\mathbf{x} | \mathbf{x} \in \mathbb{R}^n\} = \{A\mathbf{y} | \mathbf{y} \in R(A^T)\}$$